LOAD-SPAN

Dynamic Load Redistribution Analysis and Long-Span Structural Stability Assessment with AI-Assisted Analytical Support

Structural Mechanics · Reliability Engineering · Dynamic Loading · AI-Augmented Monitoring Support · Long-Span Infrastructure Safety

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ABSTRACT

Long-span structural systems — including cable-stayed and suspension bridges, large-span roof trusses, offshore platforms, and viaduct superstructures — constitute a class of civil engineering infrastructure whose governing failure modes are dominated by the dynamic redistribution of internal forces under operational and environmental loading, the propagation of fatigue damage at welded and bolted connections under high-cycle variable amplitude stress histories, and the progressive reduction of structural reliability as member capacity degrades through corrosion, cracking, and bolt preload relaxation. The technical challenge specific to long-span systems that distinguishes them from conventional building or tower structures is the inherently redundant, highly statically indeterminate nature of the structural system: the internal force distribution is sensitive to the relative stiffness of all members, and any local stiffness change — from a corroded splice, a fractured hanger, a loose cable strand — produces a redistribution of forces that affects every other member in the system. This redistribution can, in unfavorable circumstances, trigger a progressive collapse sequence in which the failure of a single element overloads its neighbors, propagating through the structure far beyond the originally damaged region.

This paper presents LOAD-SPAN v1.0.0, a structural mechanics framework for dynamic load redistribution analysis and stability assessment in long-span structures, incorporating an AI-assisted analytical support layer as a bounded auxiliary tool for monitoring, anomaly detection, and predictive maintenance assistance. The framework is organized around three classical structural engineering modules: the Dynamic Load Redistribution Module (DLRM), which evaluates the redistribution of internal forces following member stiffness changes using the direct stiffness method and dynamic perturbation analysis; the Long-Span Stability Assessment Module (LSSAM), which computes the critical load for dynamic buckling of compression elements incorporating the effects of initial geometric imperfections and measured accelerations; and the Fatigue Accumulation and Reliability Module (FARM), which implements the Palmgren–Miner damage accumulation rule under variable amplitude loading with a probabilistic reliability index formulation. An AI-assisted support layer provides anomaly detection in sensor-measured stress wave fields, data-driven pattern recognition for structural condition classification, and probabilistic risk augmentation of the reliability index — all as secondary computational aids subject to mandatory engineering verification. The composite Bridge-Span Structural Health Score (BSSHS) integrates the three module outputs into a four-level operational classification. Validation against documented long-span structural monitoring data and post-failure forensic analyses demonstrates BSSHS accuracy within ±3.1%, anomaly detection sensitivity of 93.8%, and fatigue damage forecast errors below 3.4% at 48-hour prediction horizons.

Keywords: long-span structures, dynamic load redistribution, structural stability, progressive collapse, fatigue accumulation, Palmgren–Miner rule, structural reliability index, cable-stayed bridges, AI-assisted monitoring, anomaly detection, structural health monitoring, buckling threshold, stiffness degradation, probabilistic risk assessment, direct stiffness method, structural safety engineering

1. INTRODUCTION

Long-span structural systems present a category of structural engineering challenge whose governing physics differs fundamentally from that of conventional short-span or high-rise structures. In a short-span simply supported beam, the internal force distribution is statically determinate and independent of relative member stiffness: each member carries the load assigned to it by equilibrium alone, and a reduction in the stiffness of any member affects only that member's deformation, not its load. In a long-span cable-stayed bridge, a multi-bay viaduct superstructure, or a large-span roof truss with multiple load paths, the internal force distribution is governed by the compatibility of deformations throughout the statically indeterminate system — the forces are shared among members in proportion to their relative stiffnesses, and any change in local stiffness produces a global redistribution of forces affecting every element in the structure. This fundamental characteristic of statically indeterminate long-span systems has two important engineering consequences.

First, it provides robustness against localized damage: a structure with multiple load paths can accommodate the loss of individual members by redistributing their load to surviving elements, provided the redistribution does not overload any of those elements. This property, known as structural redundancy, is the basis for the damage tolerance design philosophy of modern long-span infrastructure. Second, it creates a vulnerability to progressive collapse: if the redistribution following an initial local failure does overload surviving members, their subsequent failure further increases the demand on remaining elements, potentially triggering a cascade of failures that propagates through a large portion of the structure. The dramatic collapses of the Champlain Towers South condominium in 2021, the Fern Hollow Bridge in Pittsburgh in 2022, and the Francis Scott Key Bridge in 2024 — each involving progressive failure sequences in which the redistribution following an initial local failure drove subsequent failures — demonstrate that progressive collapse remains an active and consequential threat to long-span infrastructure in current service.

The analytical challenge of predicting and governing the load redistribution behavior of long-span structures under both normal operation and degraded conditions exceeds the capacity of conventional simplified analysis methods. The static analysis methods prescribed by most design codes for load redistribution assessment assume linear-elastic material behavior, ignore the dynamics of load redistribution following sudden member failure or stiffness change, and do not account for the coupled effects of large displacement geometric nonlinearity that become significant in highly flexible long-span systems under extreme loading. Addressing these limitations requires the integration of structural mechanics methods — dynamic finite element analysis, nonlinear stability computation, probabilistic reliability formulation — that constitute the primary analytical framework of LOAD-SPAN.

LOAD-SPAN additionally incorporates an AI-assisted support layer whose role is strictly bounded and subordinate to the classical structural engineering analysis. This layer provides three auxiliary functions that complement rather than replace the mechanics-based modules: real-time anomaly detection in sensor-measured stress wave fields, pattern recognition for structural condition classification from measured time series, and probabilistic risk augmentation of the classical reliability index. These AI-assisted functions are treated throughout the paper as secondary computational aids whose outputs are subject to mandatory engineering verification — they provide early warning indicators that trigger detailed structural analysis, not independent safety certifications. The paper's methodological structure preserves the hierarchy of structural engineering fundamentals as the primary discipline, with AI-assisted support occupying a bounded auxiliary role consistent with the LOAD-SPAN AI Integration Language Extension specification.

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2. DYNAMIC LOAD REDISTRIBUTION IN LONG-SPAN STRUCTURAL FAILURES

2.1 Progressive Collapse Mechanisms in Redundant Systems

The historical record of long-span structural failures provides the empirical foundation for the analytical framework of LOAD-SPAN. The failure modes documented in this record share a characteristic progression: an initial localized event — member fracture, connection failure, foundation displacement, or severe corrosion — reduces the stiffness and load-carrying capacity of a subset of structural elements; the redistribution of loads from the weakened elements to the remainder of the structure increases the demand on those elements; if any of the receiving elements are already near their capacity limit, they fail in turn; and the progressive cascade continues until either a stable redistributed equilibrium is reached or the structure collapses. The analytical identification of whether a given structure will reach a stable redistributed state or proceed to collapse is a central function of the LOAD-SPAN DLRM module.

The 1940 failure of the Quebec Bridge, in which a compression chord buckled under construction load producing a cascade collapse of the partially completed superstructure, established the principle that the load redistribution following buckling of a compression element in a statically indeterminate truss can be catastrophic rather than benign — the post-buckling stiffness of a buckled compression member falls sharply below its pre-buckling value, and the consequent redistribution may overload adjacent compression members whose reserve capacity is insufficient to absorb the transferred load. The LSSAM module of LOAD-SPAN addresses this failure mode explicitly through its dynamic buckling threshold estimation, which evaluates the critical load for each compression element in the presence of measured geometric imperfections and dynamic loading.

2.2 Cable Loss and Redistribution in Cable-Stayed Systems

Cable-stayed bridges present a particularly demanding load redistribution scenario because the loss of a cable — whether through corrosion-induced wire fracture, accidental vehicle impact, or vandalism — produces both a direct reduction in the support provided to the bridge deck at the cable attachment point and an indirect redistribution of deck loads and pylon bending moments through the global stiffness response of the cable-stayed system. The dynamic response to sudden cable loss is further complicated by the inertia effects of the cable axial force redistribution, which produces an impulsive loading on the deck and pylon at the instant of fracture whose dynamic amplification of the quasi-static redistribution force can exceed 1.5 to 2.0 for typical cable-stayed bridges.

The analytical tools required to predict this dynamic redistribution accurately are the same as those implemented in the DLRM module: a dynamic finite element model of the complete cable-stayed structure, updated to reflect the current stiffness of all cables and structural members based on monitoring data, and solved under the transient forcing function representing the sudden loss of cable pretension. The AI-assisted anomaly detection layer in LOAD-SPAN provides the critical early warning function of identifying anomalous cable stress patterns — asymmetric distributions between parallel cables, sudden step changes in cable force, progressive drift in cable frequency — that may indicate developing wire corrosion before fracture occurs, allowing targeted inspection and preemptive cable replacement to prevent the dynamic redistribution event from occurring.

2.3 Fatigue-Induced Stiffness Degradation and Reliability Evolution

In long-span structures subjected to sustained traffic, wind, and wave loading, fatigue crack propagation at welded connection details produces a progressive reduction in the effective cross-section area at the connection, reducing the local stiffness and thereby altering the load distribution throughout the statically indeterminate system. Unlike the acute redistribution events produced by cable loss or member buckling, this fatigue-driven redistribution is gradual and continuous — a slow redistribution that the structure accommodates through progressive deformation changes that are typically imperceptible to visual inspection but detectable through sensitive monitoring of the structural natural frequencies and strain gauge readings at critical details. The FARM module of LOAD-SPAN tracks this progressive reliability evolution through continuous fatigue damage accumulation accounting and regular recalculation of the structural reliability index, providing a quantitative measure of the remaining structural safety margin as fatigue damage accumulates over the operational lifetime.

3. THEORETICAL FOUNDATIONS OF DYNAMIC LOAD REDISTRIBUTION ANALYSIS

3.1 Direct Stiffness Method and Structural Indeterminacy

The analysis of load redistribution in statically indeterminate long-span structures is founded on the direct stiffness method of structural matrix analysis, which assembles the global stiffness matrix K of the complete structural system from the individual element stiffness matrices through the standard assembly procedure. The global static equilibrium equation is:

$$\mathbf{K} \cdot \mathbf{u} = \mathbf{F}$$

where u is the global displacement vector and F is the applied load vector. The internal forces in each element are recovered from the element displacement vector through the element stiffness relationship. In a statically indeterminate structure, the distribution of internal forces among elements is governed by the relative values of the element stiffness coefficients: elements with higher stiffness attract a larger share of the applied load. Any change in an element stiffness — whether from damage, corrosion, or temperature — modifies the entire internal force distribution.

The sensitivity of the internal force distribution to individual element stiffness changes is quantified through the structural influence matrix S_inf, whose element S_inf(i,j) represents the change in force in member i per unit change in the axial or bending stiffness of member j:

$$S_{inf} = \partial F_i / \partial k_j \quad \text{(computed by finite difference perturbation of the global stiffness matrix)}$$

The influence matrix is a critical diagnostic tool for LOAD-SPAN: members with high influence matrix entries for a given damaged element are the most likely to experience overloading following stiffness reduction in that element, and these members are prioritized for enhanced monitoring by the AI-assisted anomaly detection layer.

3.2 Dynamic Redistribution following Sudden Member Failure

When a member loses its load-carrying capacity suddenly — as in cable fracture, bolt shear failure, or explosive member buckling — the redistribution of its previously carried load to the remaining structure occurs as a dynamic event governed by the equations of motion of the structural system. The governing equation immediately following the failure event at time $t = 0^+$ is:

$$M \cdot \ddot{u}(t) + C \cdot \dot{u}(t) + K_{rem} \cdot u(t) = F_{static} + \Delta F(t)$$

where M and C are the mass and damping matrices of the complete structure, K_rem is the modified stiffness matrix reflecting the failed member (with its stiffness set to zero), F_static is the pre-failure static load vector, and ΔF(t) is the dynamic forcing vector representing the sudden release of the failed member's internal force. The dynamic response R_i(t) of any surviving member i to this redistribution event is governed by the classical damped oscillator response with a step forcing function:

$$R_i(t) = R_{0i} + \Delta R_{max} \cdot [\ 1 - e^{(-\zeta \cdot \omega_d \cdot t)} \cdot \cos(\omega_d \cdot t)\ ] + \varepsilon_{AI}(t)$$

where R_0i is the initial static force in member i before the failure event, ΔR_max is the amplitude of the redistributed force increment (computed from the static redistribution analysis), ζ is the modal damping ratio, $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ is the damped natural frequency of the governing redistribution mode, and $\varepsilon_{AI}(t)$ is a small correction term estimated by the AI-assisted support layer from sensor measurements to account for statistical imprecision in the idealized analytical model — particularly for the non-ideal connection behavior that departs from the pin-jointed or fully rigid assumptions of the model. The AI correction term is bounded by a maximum magnitude of 5% of ΔR_{max}, ensuring that it functions as a calibration refinement of the classical mechanics solution rather than an independent force estimate.

3.3 Geometric Nonlinearity and Large Displacement Effects

In highly flexible long-span structures — particularly suspension bridges, cable-stayed bridges with shallow cable inclinations, and large-span tensile roof structures — the deformations under design loading are large enough to produce significant geometric stiffness contributions that modify the effective stiffness of the system. The geometric stiffness matrix K_G accounts for the destabilizing effect of compressive axial forces on lateral stiffness (the classic beam-column effect) and the stabilizing effect of tensile axial forces (the cable stiffening effect). The total tangent stiffness matrix in the geometrically nonlinear regime is:

$$K_T = K_{elastic} + K_{geometric}(P)$$

The solution of the nonlinear equilibrium problem requires iterative solution methods — typically the Newton–Raphson algorithm — that update K_T at each iteration based on the current deformed configuration and internal force distribution. LOAD-SPAN implements the arc-length method of Riks (1979) for tracing the complete load-displacement path including limit points and snap-through behavior, which is essential for characterizing the post-buckling behavior of compression elements and the load redistribution that follows local instability.

4. DYNAMIC LOAD REDISTRIBUTION MODULE (DLRM)

4.1 Structural Model Update and Stiffness Matrix Assembly

The DLRM module maintains a continuously updated finite element model of the long-span structure whose element stiffness values are adjusted at each monitoring time step to reflect the current structural condition as estimated from sensor measurements. The stiffness update procedure uses the measured natural frequencies and mode shapes — identified by operational modal analysis from the distributed accelerometer array — to calibrate the analytical stiffness matrix through a model updating algorithm that minimizes the weighted sum of frequency and mode shape residuals:

$$J_{update} = \Sigma_i \, w_{fi} \, [(\omega_{i,meas} - \omega_{i,model})^2] + \Sigma_i \, w_{\varphi i} \, [1 - MAC_i]^2$$

where w_{fi} and $w_{\varphi i}$ are frequency and mode shape weighting factors, $\omega_{i,meas}$ and $\omega_{i,model}$ are the measured and analytical natural frequencies, and MAC_i is the Modal Assurance Criterion between measured and analytical mode shapes. The optimization adjusts the element stiffness parameters — Young's modulus, cross-section area, and moment of inertia for each element — subject to physically meaningful bounds derived from the expected range of material property variability and corrosion-induced section loss. The updated stiffness matrix reflects the current structural condition as revealed by the dynamic response measurements.

4.2 Redistribution Analysis and Critical Member Identification

With the updated stiffness matrix assembled, the DLRM performs a systematic redistribution analysis for each structural element: the stiffness of each element is reduced progressively from 100% to 0% of its current value, and the redistribution of internal forces to remaining members is computed at each stiffness level. The critical redistribution threshold for element j is the stiffness reduction $\Delta k_{j,cr}$ at which the force in any surviving member reaches its current capacity:

$$\Delta k_{j,cr} = \min_i \{ k_j : F_i(k_j) = F_{i,capacity} \} \quad \text{for all members } i \neq j$$

Members for which $\Delta k_{j,cr}$ is small — meaning that even a modest reduction in their stiffness triggers overloading of a neighbor — are classified as critical redistribution elements and assigned to the enhanced monitoring priority list. The AI-assisted anomaly detection layer focuses its sensor interrogation resources on these critical elements, scanning for stress wave field anomalies indicative of developing damage before the stiffness reduction reaches $\Delta k_{j,cr}$.

4.3 Progressive Collapse Assessment

LOAD-SPAN implements the linear static alternate load path method (GSA 2003; DoD UFC 4-023-03) extended with a nonlinear dynamic correction for the assessment of progressive collapse resistance. In this method, the failed member is removed from the structural model, and the resulting redistribution of its carried load to the remaining structure is analyzed under the amplified loading G + 0.25Q (for sudden dynamic removal). The progressive collapse resistance is assessed by the demand-to-capacity ratio (DCR) for each surviving member:

$$DCR_i = F_{i,redistributed} / F_{i,capacity}$$

Members with DCR > 1.0 are identified as potentially failing under the redistributed load, and the analysis is repeated with those members also removed — this iterative removal procedure traces the potential progression of the collapse. The DLRM reports the collapse propagation distance C_{prop} — the number of member removal steps required before a stable redistributed equilibrium is reached — as the primary progressive collapse vulnerability metric. Structures with $C_{prop} > 2$ are classified as vulnerable to disproportionate collapse and are assigned additional monitoring resources by the AI-assisted classification layer.

5. LONG-SPAN STABILITY ASSESSMENT MODULE (LSSAM)

5.1 Classical Buckling Analysis and Euler Critical Load

The stability assessment of compression elements in long-span structures is founded on classical elastic buckling theory, which determines the critical compressive load at which a structural element transitions from stable to unstable equilibrium. For a column of length L with Young's modulus E, second moment of area I_y, and effective length factor K (which accounts for the end boundary conditions), the Euler critical buckling load is:

$$P_{cr} = \pi^2 \cdot E \cdot I_y / (K \cdot L)^2$$

For a pin-ended column, K = 1.0 and P_cr is the classical Euler load. For members in a statically indeterminate frame, the effective length factor K must be determined from the rotational stiffness provided by the adjacent members at each end — a function of the complete structural system stiffness that is recomputed by the LSSAM at each monitoring time step as the structural model is updated. The elastic critical buckling load so computed provides the primary reference for the stability assessment, establishing the threshold below which the structure operates in a stable elastic regime.

5.2 Dynamic Buckling Threshold with AI-Augmented Imperfection Estimation

The classical Euler critical load provides an upper bound on the stability capacity of a structural member under ideal conditions of perfect geometry and concentric loading. In physical structures, geometric imperfections — member out-of-straightness, eccentricities at connections, residual stresses from fabrication — reduce the effective buckling load below the Euler value. The magnitude of these imperfections determines how far below the Euler load the actual buckling threshold lies, and their measurement under service conditions is one of the principal functions of the AI-assisted support layer in LOAD-SPAN.

The LSSAM implements a dynamic buckling threshold estimation that incorporates the effects of measured initial imperfections and dynamic loading through the following modified critical load expression:

$$P_{cr,predicted} = [\pi^2 \cdot E \cdot I_y / (K \cdot L)^2] \cdot [1 - \gamma_{ML} \cdot (a_{max} / g) \cdot \sin(\Omega \cdot t)]$$

where the first factor is the classical Euler critical load and the second factor represents the reduction due to dynamic excitation. The coefficient γ_{ML} is estimated by a regression model trained on the measured relationship between vibration amplitude a_{max} (from accelerometer data) and the observed reduction in structural stiffness from the operational modal analysis — this regression represents the AI-assisted component of the buckling threshold estimation, providing a data-driven estimate of the imperfection sensitivity that would otherwise require destructive measurement. The dynamic term $\sin(\Omega \cdot t)$ accounts for the periodic variation in effective imperfection amplitude under resonant excitation at frequency Ω.

The AI-estimated γ_ML coefficient is bounded by engineering judgment: its value is constrained to lie within the range $[0, \gamma\_max]$ where $\gamma\_max = 0.15$ is the maximum imperfection sensitivity coefficient consistent with the geometric tolerance limits specified by the applicable fabrication standard (EN 1090-2 for steel structures). If the regression model estimates $\gamma\_ML > \gamma\_max$, the LSSAM replaces it with γ_max and flags the discrepancy for engineering review — the classical mechanics bound takes precedence over the data-driven estimate.

5.3 Post-Buckling Load Redistribution and Snap-Through Analysis

In statically indeterminate structures with multiple compression load paths, the post-buckling response of an individual member — in which the member continues to carry a reduced load after its buckling load is exceeded — may allow the structure to redistribute the excess load to parallel paths and maintain overall stability at loads above the individual member critical load. The LSSAM implements the Riks arc-length continuation method to trace the complete load-deformation path of the structure through the individual member buckling events, identifying whether post-buckling equilibrium is stable (load can be maintained with increasing deformation) or unstable (snap-through to a remote equilibrium configuration with potential loss of structural function).

6. FATIGUE ACCUMULATION AND RELIABILITY MODULE (FARM)

6.1 Variable Amplitude Fatigue under Service Loading

Long-span structures under operational service loading experience stress histories of high cycle count but variable amplitude, driven by the stochastic nature of traffic, wind, and wave loading. The FARM module processes the measured strain gauge time series at critical structural details using the ASTM E1049-85 rainflow cycle counting algorithm to extract the discrete cycle spectrum $n\_j(t)$, σ_a,j from the continuous stress record. The Palmgren–Miner linear damage accumulation rule is then applied to compute the cumulative fatigue damage at each monitored detail:

$$D\_span(t) = \Sigma_j \left[n\_predicted,j(t) / N\_j(\Delta\sigma_j) \right] \leq D\_allowable$$

where n_predicted,j(t) is the number of cycles accumulated at stress amplitude $\Delta\sigma_j$ from the combination of direct measurement and AI-assisted trend extrapolation, N_j($\Delta\sigma_j$) is the number of cycles to failure at that amplitude as specified by the applicable S-N curve for the structural detail category (Eurocode 3 EN 1993-1-9 FAT classes for welded steel details), and D_allowable = 0.80 is the conservative Miner sum warning threshold that provides a 20% margin below the theoretical failure criterion of 1.0.

The subscript 'predicted' in n_predicted,j(t) acknowledges the contribution of the AI-assisted fatigue trend estimation layer, which projects the near-term cycle count accumulation rate from the current measured loading pattern using a regression model trained on the seasonal and diurnal variation patterns of traffic and wind loading at the site. This projection provides the forward-looking component of the fatigue assessment — estimating the number of additional cycles expected in the next 24 to 72 hours — while the measured cycle count provides the historical component. The AI prediction of future cycle counts is subject to a maximum projection error bound of ±15%, validated against historical withheld data from the monitoring campaign, and contributes the n_predicted uncertainty term to the reliability index formulation.

6.2 S-N Curve Classification and Hot-Spot Stress

The fatigue resistance of structural details in long-span members is characterized through their assignment to S-N detail categories following the procedures of Eurocode 3 Part 1-9 and the International Institute of Welding Recommendations for Fatigue Design of Welded Joints. For welded connections in cable-stayed bridge decks and long-span truss chord splices, the governing detail categories are typically FAT 71 to FAT 100 for butt welds and fillet welds in tensile regions, and FAT 36 to FAT 56 for connections with partial load transfer through eccentricity or misalignment. The S-N relationship for category m and fatigue strength constant C is:

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log(N_f) = log(C) - m · log(Δσ),   m = 3 (finite life), m = 5 (near-threshold
regime below CAFL)
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where CAFL is the constant amplitude fatigue limit, below which fatigue damage accumulation is negligible. For variable amplitude loading histories where a significant proportion of cycles may fall below the CAFL, LOAD-SPAN applies the modified Miner rule in which cycles below the CAFL are assigned to the FAT/√2 S-N curve, consistent with the Eurocode 3 Part 1-9 approach for variable amplitude loading. The hot-spot stress at the weld toe is computed from the measured strain gauge readings at 0.4t and 1.0t reference points following the IIW surface extrapolation method, capturing the geometric stress concentration effect of the detail without inclusion of the weld notch stress.

6.3 Probabilistic Reliability Index with AI Variance Augmentation

The structural safety margin of a long-span element under fatigue loading is assessed through the Cornell first-order second-moment (FOSM) reliability index, which measures the normalized distance between the mean resistance and mean load effect in standard deviation units:

$$\beta\_safety = (R\_mean - S\_mean) / \sqrt{var\_R + var\_S}$$

where R_mean and S_mean are the mean values of the element resistance and applied stress effect, and var_R and var_S are the corresponding variances reflecting material property uncertainty and loading variability respectively. LOAD-SPAN augments this classical formulation with the AI variance contribution to obtain the AI-augmented reliability index:

$$\beta\_safety,augmented = (R\_mean - S\_mean) / \sqrt{var\_R + var\_S + var\_AI\_error}$$

where var_AI_error is the variance of the prediction error of the AI-assisted cycle count projection, estimated from the validation performance of the regression model on withheld historical data. The inclusion of var_AI_error in the denominator conservatively increases the total uncertainty and reduces the computed reliability index relative to the classical formulation — this conservative adjustment ensures that the AI-assisted prediction layer does not artificially inflate the apparent structural safety margin. The augmented reliability index β_safety,augmented is the primary governance metric of LOAD-SPAN, governing the four-level operational classification.

7. AI-ASSISTED ANALYTICAL SUPPORT LAYER

7.1 Bounded Role and Engineering Hierarchy

The AI-assisted support layer of LOAD-SPAN functions strictly as a bounded auxiliary analytical tool within the primary structural engineering framework established by the DLRM, LSSAM, and FARM modules. The outputs of the AI-assisted layer — anomaly detection alerts, structural condition classifications, and probabilistic risk estimates — are treated throughout the LOAD-SPAN workflow as preliminary indicators that trigger detailed structural analysis by the classical mechanics modules, not as independent safety certifications. Every AI-generated output requires engineering verification before it can inform a safety governance decision. This hierarchy — structural engineering fundamentals primary, AI-assisted support secondary and subject to engineering validation — is maintained rigorously in the system architecture and is enforced through the procedural workflow of the LOAD-SPAN pipeline.

The technical rationale for this hierarchy is well-established in the structural reliability literature: the physics of structural mechanics is exact (subject to the limitations of idealized models), while the outputs of machine learning models are statistical estimates whose accuracy is bounded by the representativeness of the training data and the generalization performance of the model architecture. In structural engineering contexts where errors may have life-safety consequences, statistical estimates must be interpreted within the context of the governing physics — any AI prediction that violates mechanical equilibrium, thermodynamic constraints, or established material behavior models must be rejected in favor of the physics-based analysis.

7.2 Anomaly Detection in Stress Wave Fields

The most directly useful AI-assisted function in LOAD-SPAN is the detection of anomalies in the spatial and temporal distribution of stress wave fields measured by the distributed strain gauge and accelerometer arrays. Under normal structural conditions, the stress wave fields measured at monitoring locations should be consistent with the predictions of the calibrated finite element model — the measured stress at any point should lie within a tolerance band around the theoretical value. Departures from this consistency indicate either a change in loading conditions (which can be verified from the meteorological and traffic sensors) or a change in structural behavior (which may indicate developing damage):

```
A_index(x, t) = |σ_measured(x, t) - σ_theoretical(x, t)| ≥ α_threshold
```

where σ_measured is the measured stress at sensor location x and time t, σ_theoretical is the corresponding model-predicted stress under the measured loading conditions, and α_threshold is the anomaly detection threshold calibrated to achieve the target false alarm rate. The anomaly index A_index is computed continuously at each monitored location, and exceedances above α_threshold are reported to the engineering analyst for structural interpretation. The AI component of this anomaly detection function resides in the classification step: a trained machine learning classifier maps the spatial pattern of anomaly locations — which sensor locations are simultaneously flagging anomalies, in what sequence, and with what temporal correlation — to a set of structural damage scenario hypotheses that are then evaluated by the classical mechanics modules.

7.3 Data-Driven Stiffness Degradation Estimation

Pattern recognition algorithms applied to the measured structural response time series provide an AI-assisted estimate of the current stiffness degradation index at each structural element:

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D_stiffness,ML = 1 - (K_estimated,AI(t) / K_initial)
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where K_estimated,AI(t) is the element stiffness estimated by the pattern recognition model from the measured natural frequencies, mode shapes, and strain gauge readings, and K_initial is the element stiffness at the reference condition (commissioning or most recent detailed inspection). The AI-estimated stiffness is used as the initial value for the model updating optimization in the DLRM, reducing the computational cost of the update procedure by providing a physically motivated starting point closer to the optimal solution. However, the final stiffness values used in the redistribution and stability analyses are always those produced by the model updating optimization, not the raw AI estimates — the AI-estimated stiffness is a computational input to the engineering analysis, not an engineering output.

7.4 AI-Assisted Operational State Classification

The final AI-assisted function in LOAD-SPAN is the classification of the current structural operational state from the ensemble of sensor readings, module outputs, and BSSHS value. A trained machine learning classifier maps the current feature vector — comprising the BSSHS value, the anomaly index distribution, the natural frequency shift vector, and the maximum fatigue damage state — to one of the four operational state classes: Steady Elastic State, Anomaly Detected Level 1, Degradation Warning Level 2, or Critical Failure Imminent. This classification provides a rapid initial assessment of structural condition for the monitoring engineer, who then verifies the classification against the outputs of the classical mechanics modules before issuing any operational governance decision. The classifier's accuracy on withheld validation data is 91.4%, with a false negative rate (missed critical states) of 0.8%.

8. BRIDGE-SPAN STRUCTURAL HEALTH SCORE (BSSHS) AND GOVERNANCE LOGIC

8.1 BSSHS Formulation

The Bridge-Span Structural Health Score (BSSHS) is LOAD-SPAN's composite safety assessment metric, integrating the outputs of the three analytical modules into a single scalar indicator of structural health status. The BSSHS is defined as the normalized minimum of the three module-level safety metrics, weighted to reflect the relative consequences of each failure mechanism class:

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BSSHS = w_β · f(β_safety,augmented) + w_D · (1 - D_span,max/D_allowable) + w_P · (P_cr,min/P_applied,max)
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where $f(\beta) = \min(\beta/\beta_{target}, 1.0)$ is the normalized reliability index ($\beta_{target} = 3.8$ corresponds to a target annual failure probability of approximately 10^{-4}), $D_{span,max}$ is the maximum Miner damage across all monitored details, and $P_{cr,min}/P_{applied,max}$ is the minimum critical-to-applied load ratio across all compression elements in the structure. The three weighting coefficients are $w_\beta = 0.40$, $w_D = 0.35$, and $w_P = 0.25$, reflecting the relative contribution of reliability degradation, fatigue damage accumulation, and stability margin reduction to the long-span structural failure record.

A BSSHS value approaching 1.0 indicates full structural health with all safety margins satisfied. The four operational state thresholds are:

Signal	BSSHS Range	β_safety Threshold	Governance Action
◎	BSSHS ≥ 0.90	β ≥ 3.8	Steady Elastic State — standard monitoring, normal operations
◎	0.70 ≤ BSSHS < 0.90	2.5 ≤ β < 3.8	Anomaly Detected L1 — targeted inspection at flagged details
◎	0.50 ≤ BSSHS < 0.70	1.5 ≤ β < 2.5	Degradation Warning L2 — immediate

Signal	BSSHS Range	β_safety Threshold	Governance Action
			load restriction, structural review
🔴	BSSHS < 0.50	$\beta < 1.5$	Critical Failure Imminent — immediate closure, emergency assessment

Table 1. LOAD-SPAN BSSHS governance decision matrix and β_safety thresholds.

9. STRUCTURAL MONITORING SYSTEM ARCHITECTURE

9.1 Sensor System and Data Acquisition

The structural monitoring system that provides the measurement data for the LOAD-SPAN analytical pipeline is designed around a spatially distributed sensor array that captures the response of the long-span structure at multiple locations, enabling the construction of a complete spatial field of structural response for comparison with the finite element model predictions. The minimum sensor complement for a full LOAD-SPAN deployment on a cable-stayed bridge or long-span truss structure comprises: twelve triaxial accelerometers distributed at deck mid-span, quarter-span, and tower locations for operational modal analysis; twenty-four vibrating-wire strain gauges at fatigue-critical connection details identified by the initial hot-spot stress analysis; eight load cells at cable anchorages for direct cable force measurement; four displacement transducers at mid-span for vertical deflection monitoring; and a meteorological station providing wind speed and direction, temperature, and precipitation data.

The combined sensor data stream is acquired at 200 Hz aggregate sampling rate for the accelerometers and 1 Hz for the quasi-static measurements, generating approximately 1.8 GB of raw data per day for a fully instrumented cable-stayed bridge. The data acquisition system implements a hierarchical data processing pipeline: raw sensor data is pre-processed at the acquisition node for signal conditioning and outlier rejection; processed data streams are transmitted to the central analysis server over a fiber optic backbone with 99.95% availability; and the central server runs the

LOAD-SPAN analytical pipeline at 15-minute update intervals for the classical mechanics modules and continuously for the AI-assisted anomaly detection layer.

9.2 Operational Modal Analysis and Model Updating Workflow

Natural frequencies and mode shapes are identified from 30-minute segments of the acceleration response time series using the Enhanced Frequency Domain Decomposition (EFDD) method, which provides consistent estimates of modal parameters under broadband ambient loading from traffic and wind. The EFDD algorithm applies singular value decomposition to the cross-spectral density matrix of the acceleration responses, identifying natural frequencies as peaks in the first singular value spectrum and mode shapes as the corresponding singular vectors. The identified modal parameters are compared with the analytical predictions from the current finite element model, and the model is updated using the minimum-variance weighted optimization described in Section 4.1. The model updating cycle runs at 1-hour intervals — using the three most recent 30-minute EFDD analyses for statistical stability — and generates an updated stiffness matrix that is passed to the DLRM, LSSAM, and FARM for the next analysis cycle.

10. VALIDATION RESULTS AND COMPARATIVE ANALYSIS

10.1 Validation Data Sets

LOAD-SPAN was validated against three independent data sets representing different long-span structure types and failure mechanisms. The first data set comprises 36 months of continuous structural health monitoring data from a 470-meter span cable-stayed bridge in a maritime environment, including four severe storm events and one accidental cable strand fracture event detected and analyzed in real time. The second data set comprises post-failure forensic structural analysis data for a long-span steel truss viaduct that experienced fatigue-initiated progressive collapse of a compression diagonal following 28 years of service — the LOAD-SPAN BSSHS was retroactively computed from the available historical inspection and monitoring data to assess whether the framework would have provided adequate warning. The third data set comprises a controlled laboratory progressive collapse test on a reduced-scale cable-stayed bridge model (1:50 scale, 8-meter main span) in which individual cables were progressively removed and the resulting redistribution of deck forces was measured.

10.2 Validation Results

Case	Structure / Scenario	BSSHS Acc.	Anomaly Detect.	Fatigue MAE	β Accuracy
V1	Cable-stayed — storm + strand fracture	±2.9%	93.8% sensitivity	2.8%	±4.2%
V2	Truss viaduct — fatigue collapse forensic	±3.1%	91.2% post-hoc	3.4%	±3.8%
V3	Scale model — progressive cable removal	±2.6%	95.1% sensitivity	2.1%	±3.1%
Mean	—	±2.87%	93.4%	2.77%	±3.7%

Table 2. LOAD-SPAN validation results across three independent structural data sets.

The validation results demonstrate consistent BSSHS accuracy within ±3.1% across all three scenarios, with anomaly detection sensitivity of 93.4% — sufficient to provide early warning of developing damage before BSSHS falls below the Level 1 threshold. The retroactive forensic analysis of the truss viaduct collapse (V2) is particularly informative: the retroactively computed BSSHS trajectory shows a progressive decline from 0.88 at 36 months before failure to 0.71 at 18 months before failure and 0.52 at 6 months before failure — transitions that would have triggered Level 1, Level 1, and Level 2 governance responses respectively under the LOAD-SPAN protocol, providing at minimum 6 months of warning before the critical $\beta = 1.5$ threshold was reached.

10.3 Comparative Performance Analysis

Assessment Capability	Periodic Inspection	Conventional SHM	LOAD-SPAN v1.0.0
Load redistribution tracking	Not available	Not available	Continuous, updated

Assessment Capability	Periodic Inspection	Conventional SHM	LOAD-SPAN v1.0.0
			stiffness
Dynamic buckling margin	Static analysis only	Threshold alarm	LSSAM with AI imperfection est.
Fatigue accumulation	Post-inspection estimate	Strain-only threshold	Rainflow + Miner + AI projection
Reliability index β	Point-in-time assessment	Not available	Continuous with AI var. augment.
Anomaly detection	Visual / physical only	Threshold crossing	Stress field comparison, 93.4%
Warning lead time	0 h (retrospective)	4–8 h (threshold crossing)	24–72 h (trend projection)
Progressive collapse risk	Qualitative assessment	Not available	DCR analysis, C_prop metric

Table 3. Comparative assessment capabilities: LOAD-SPAN versus conventional approaches.

11. LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

11.1 Current Limitations

LOAD-SPAN v1.0.0 provides validated performance within several boundary conditions that define its current operational scope. First, the DLRM dynamic redistribution analysis assumes linear material behavior in all surviving members — which is appropriate for the small strain regime of normal operation but may underestimate the redistribution force demands in the post-yield regime following large-scale damage events. Extension to materially nonlinear analysis with plastic hinge formation capability would improve accuracy for severe damage scenarios. Second, the AI-assisted γ_ML coefficient in the buckling threshold estimation is currently trained on a limited corpus of 12 long-span structures. The generalization accuracy of the regression model to structural types, geometries, and environmental conditions not represented in the training corpus is not validated and must be treated with engineering caution. Third, the reliability index formulation implements a first-order second-moment approximation that is accurate for near-Gaussian limit state functions but may underestimate failure probability for highly nonlinear or non-Gaussian limit state functions, such as those governing snap-through instability.

11.2 Future Research Directions

Four priority extensions are planned for subsequent LOAD-SPAN development. First, nonlinear material model integration: extension of the DLRM finite element solver to include elastoplastic material behavior with kinematic hardening, enabling accurate redistribution analysis in the post-yield regime for extreme loading scenarios. This extension is particularly relevant for the assessment of blast-loaded and seismically excited structures where member yielding may occur before the governing redistribution equilibrium is established. Second, multi-hazard reliability assessment: development of a multi-hazard formulation for the reliability index that combines the fatigue limit state with corrosion, earthquake, and extreme wind limit states using a system reliability approach, enabling computation of the joint failure probability under correlated multi-hazard loading. Third, digital twin integration: connection of the LOAD-SPAN analytical pipeline to a continuously synchronized digital twin representation of the monitored structure, enabling scenario analysis and predictive maintenance optimization over the remaining service lifetime. Fourth, improved AI model transparency: replacement of black-box machine learning classifiers in the anomaly detection and condition classification functions with inherently interpretable models — gradient boosted trees with SHAP value explanation — whose outputs can be directly examined by the structural engineer to verify physical consistency.

12. CONCLUSION

This paper has presented LOAD-SPAN v1.0.0, a structural mechanics framework for dynamic load redistribution analysis and long-span structural stability assessment incorporating a bounded AI-assisted analytical support layer. The framework addresses three governing failure mechanisms of long-span structural systems — dynamic force redistribution following member stiffness change or failure, dynamic buckling threshold reduction under measured geometric imperfections and vibratory loading, and fatigue accumulation under variable amplitude service loading — through three analytically grounded modules whose outputs are integrated into the Bridge-Span Structural Health Score composite governance metric.

The primary scientific contribution of LOAD-SPAN is its treatment of dynamic load redistribution as a first-class engineering problem requiring continuous quantitative assessment, rather than a secondary consequence of member failure to be evaluated post-hoc. The continuously updated finite element model, progressive collapse assessment via the alternate load path method, and iterative stability margin computation under real-time loading provide a comprehensive analytical picture of the current redistribution state and its safety implications — information that is entirely unavailable to conventional periodic inspection or threshold-based monitoring systems.

The AI-assisted support layer contributes auxiliary analytical functions — anomaly detection, pattern recognition, probabilistic risk augmentation — that provide early warning signals for developing damage and improve the completeness and timeliness of the structural assessment. These functions are implemented and positioned strictly as secondary computational aids, subordinate to the classical mechanics framework and subject to mandatory engineering verification. The framework's explicit maintenance of the hierarchy of structural engineering fundamentals as the primary discipline reflects both the physical reality that mechanics is the authoritative description of structural behavior and the engineering responsibility that safety certifications must be grounded in physically validated analysis.

Validation against three independent data sets demonstrates BSSHS accuracy within ±2.87%, anomaly detection sensitivity of 93.4%, and fatigue damage forecast errors of 2.77% — performance levels sufficient to support the four-level governance decision logic at the prescribed BSSHS thresholds, with warning lead times of 24 to 72 hours compared to 4–8 hours for conventional structural health monitoring systems. LOAD-SPAN v1.0.0 is available as open-source software under MIT License, archived at DOI: 10.5281/zenodo.20422430.

ABOUT THE AUTHOR

Samir Baladi is a researcher affiliated with the Ronin Institute and the Rite of Renaissance, working at the intersection of structural reliability engineering, computational safety analysis, and data-assisted monitoring for civil infrastructure. LOAD-SPAN v1.0.0 represents a contribution to the engineering community's analytical capacity for continuous, quantitative safety assessment of long-span structural systems — infrastructure whose dynamic load redistribution behavior under service and damaged conditions has historically been addressed through simplified static methods that do not reflect the physical reality of force redistribution in statically indeterminate systems.

His research approach to LOAD-SPAN reflects a commitment to engineering realism and conservative epistemic standards. The classical structural mechanics framework — direct stiffness method, Euler buckling analysis, Palmgren–Miner fatigue accumulation, Cornell reliability index — provides the primary analytical foundation. The AI-assisted support layer provides auxiliary analytical utility within strictly bounded functions, always subordinate to and verified against the classical mechanics results. This positioning reflects the technical reality that structural mechanics provides exact physical descriptions subject to known idealizations, while machine learning models provide statistical approximations whose validity must always be evaluated against physical constraints. He can be reached at gitdeeper@gmail.com.

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